Lab 7 – Fuzz Testing

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Objective

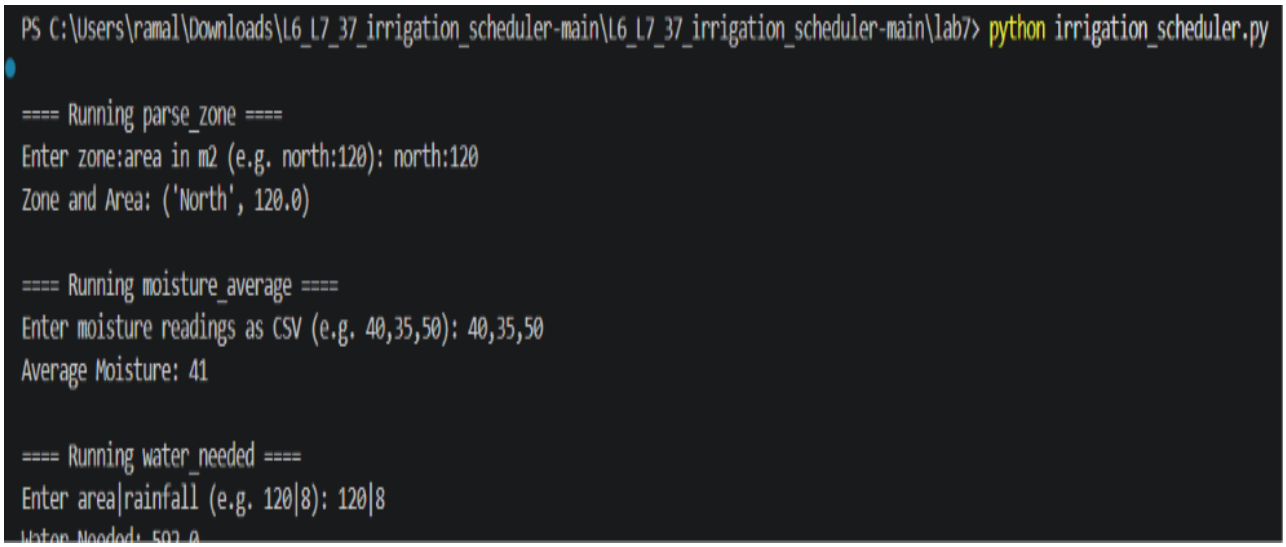
The objective of this lab is to use Hypothesis-based fuzz testing to check the irrigation scheduler functions with a range of inputs, identify failures, fix the application, and rerun the tests.

Tested functions

1. parse_zone(text)
2. moisture_average(readings_csv)
3. water_needed(text)

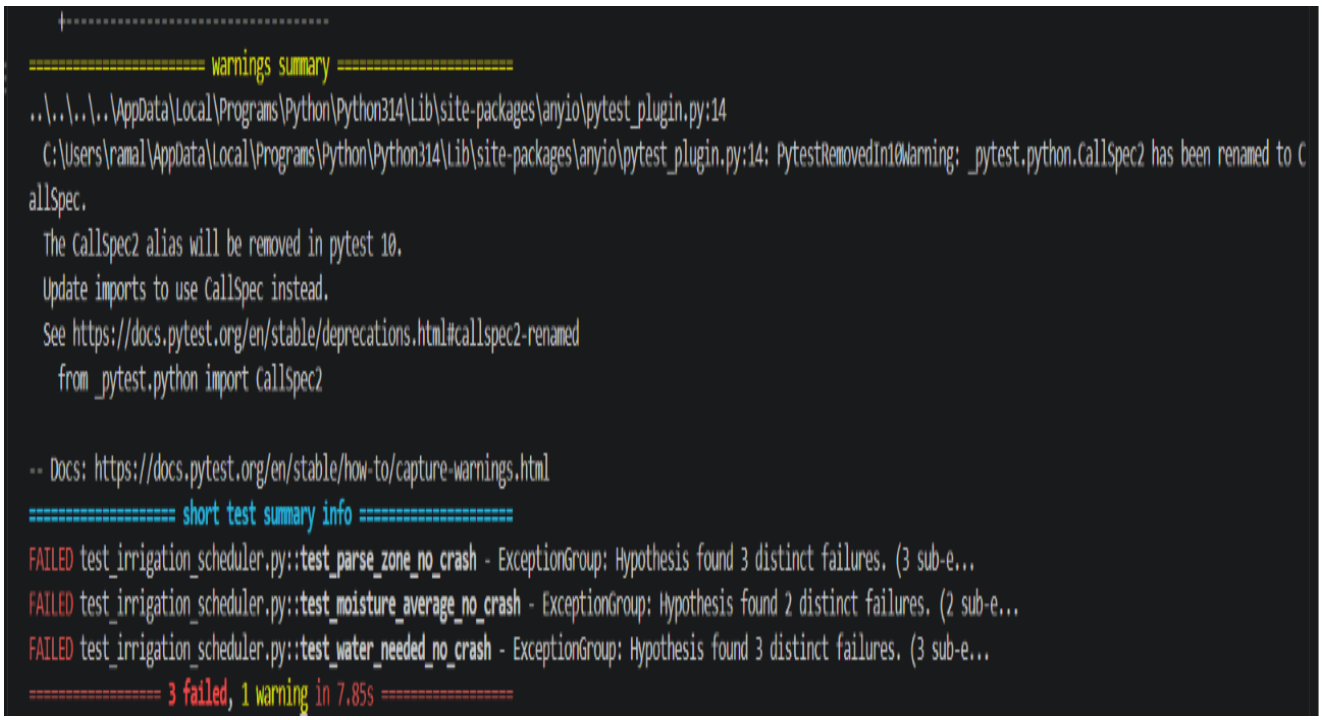
Screenshot 1 – Original application output

The original application was run with valid sample inputs before fixing the bugs.



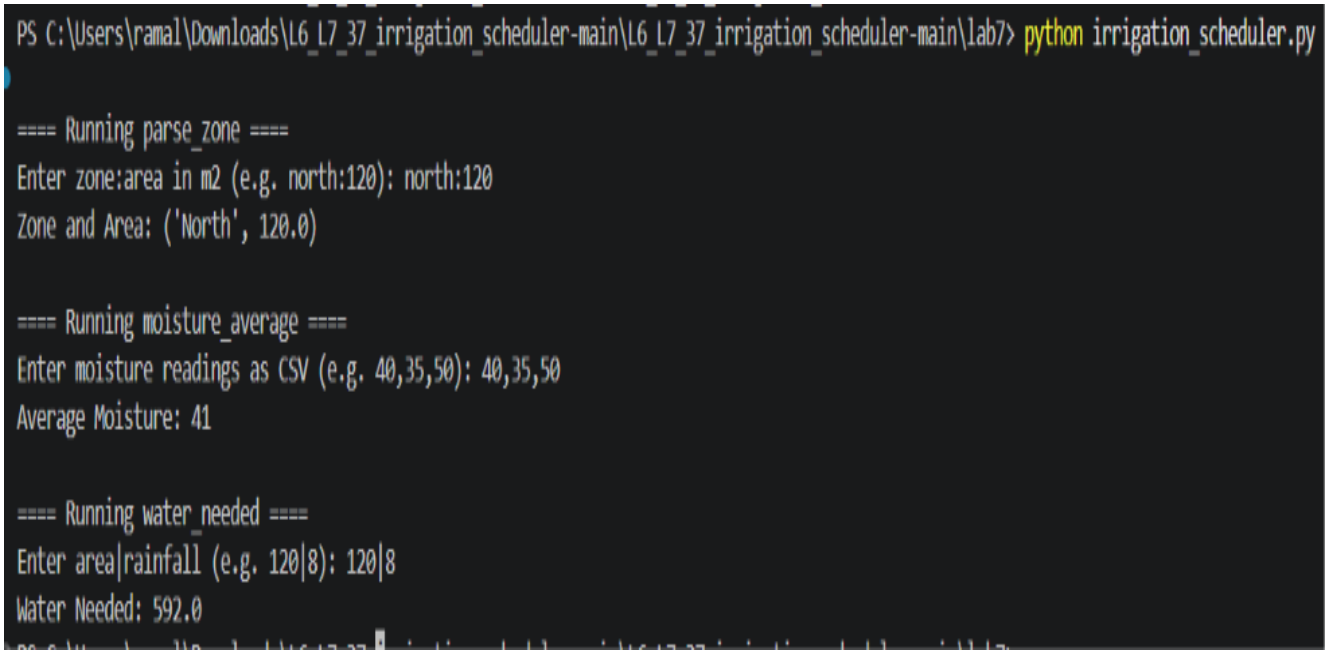
Screenshot 2 – Failing Hypothesis tests

The initial fuzz tests exposed input-handling failures in the three functions



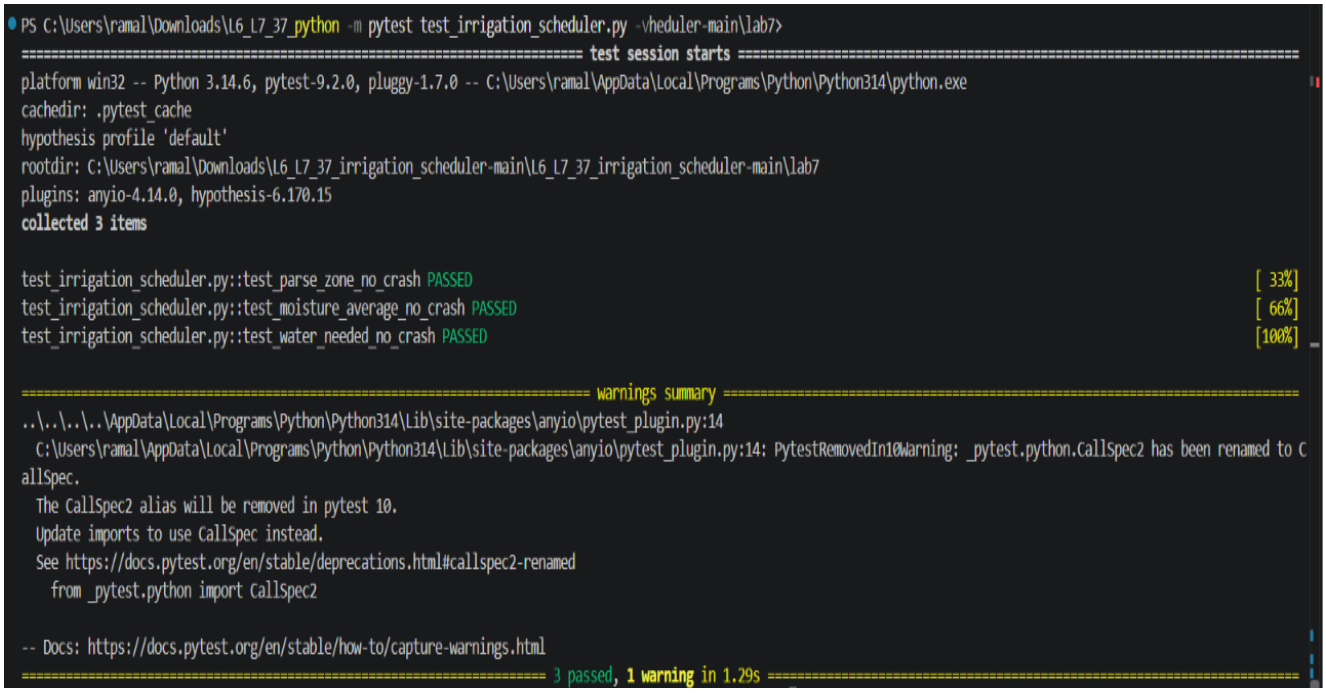
Screenshot 3 – Fixed application output

After adding input validation, the application was run again with the same valid sample inputs



Screenshot 4 – Passing Hypothesis tests

The tests were rerun after the fixes. All three tests passed. Pytest displayed one dependency deprecation warning, but it did not cause a test failure



Conclusion

Hypothesis fuzz testing identified cases that caused the original irrigation scheduler functions to raise exceptions.
The functions were updated to validate inputs safely. After the changes, all three no-crash tests passed.